Yangruirui ZHOU

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EDUCATION

Boston University, Boston, MA 09/2020-01/2026

Doctor of Philosophy (Ph.D.) in Computer Engineering

GPA: 3.91/4.00

Advisor: Douglas Densmore

Thesis: AI-driven design automation for microfluidics and multicellular systems

Research Focus: AI-enabled microfluidic CAD, formal verification, and multicellular biological system design.

University of Electronic Science and Technology of China (UESTC), Chengdu, China 09/2016-06/2020

Bachelor of Engineering in Software Engineering *(Elite program)*

GPA: 3.94/4.00 (Rank: 6/740)

Thesis: Analysis of Postoperative Applications of Pose Tracking Algorithms.

AWARDS & HONORS

iGEM "**Gold medal**" (Student Mentor, wiki: https://2024.igem.wiki/bostonu/) 11/2024

China Youth Science and Technology Innovation Award (Nominated by UESTC) 05/2020

Most Outstanding Students of UESTC 2019 (成电杰出学生, **10/5000** in UESTC, **1/740** in department) 12/2019

iGEM "**Gold medal**" and "**Best Software Project**" (wiki: https://2019.igem.org/Team:UESTC-Software) 11/2019

Outstanding Graduates Award of UESTC 10/2019

"Wu Liang Ye" Enterprise Scholarship (**2/740** in Software Engineering department) 09/2019

TECHNICAL SKILLS

AI & Data Science: Graph algorithm library (NetworkX), ML/AI/DS library (PyTorch, TensorFlow, NumPy)

Systems & Software: Python, C/C++, Java, TypeScript, JavaScript, HTML, Linux Shell, LaTeX

Biological & Microfluidic Fabrication: CNC Milling, Laser Cutting

Design & Modeling Tools: Fusion 360, Microfluidic Devices Manual Design

RESEARCH EXPERIENCE

Research Assistant | Boston University, CIDAR lab Boston, MA | 09/2020-Present

Supervisor: Prof. Douglas Densmore

- **Neptune — Hierarchical Microfluidic Design Automation System *(CUHK collaboration)***
  - Developed a full-stack microfluidic design automation system enabling end-to-end synthesis from high-level behavioral specifications to physical layout generation.
  - Developed a DSL and compiler that transform behavioral design inputs into structured connectivity graphs, which are further processed by external placement and routing algorithms; the DSL is currently supported by all five major mainstream large language models for assisted program synthesis.
  - Built a web-based interface tightly integrated into the lab's existing 3DuF design environment, enabling direct visualization, interactive editing, and manual refinement of synthesized microfluidic designs.
- **Oriole — Genetic Circuit Partitioning for Multicellular Computation *(MIT collaboration)***
  - Developed a sub-graph partitioning algorithm enabling implementation of complex computational functions (e.g., cryptographic hashing) in multicellular systems under biological and hardware constraints.
  - Designed a constraint formalization layer that encodes biological and experimental constraints into computational representations, enabling automated design and optimization in genetic circuit synthesis.
- **Vespa — In-silico Logic-level Constraint-based Validation for Large-scale Microfluidic Systems**
  - Developed an automated in-silico validation system for microfluidic workflows, enabling simulation-based verification and reliability testing prior to wet-lab execution.
  - Constructed benchmark datasets and logic-level constraint representations for evaluating and optimizing

microfluidic design automation methods.

- **Other Works**
  - Designed and fabricated modular microfluidic systems supporting multi-modal experimental workflows for studying cell–cell communication in controlled environments.
  - Built a natural language interface using large language models (Claude, GPT, Gemini, Qwen, DeepSeek) enabling users to interact with microfluidic design workflows without programming.

iGEM Team Student Mentor | BostonU (Wiki: https://2024.igem.wiki/bostonu/) Boston, MA | 05-10/2024

- Led computational and software design for iGEM 2024 project, bridging biological objectives with system-level modeling and implementation.
- Mentored students in system design and problem decomposition for biological engineering workflows.

iGEM Team Member | UESTC (Wiki: https://2019.igem.org/Team:UESTC-Software) Chengdu, China | 05-10/2019

- Built a cross-database sequence-to-protein mapping pipeline improving annotation between iGEM Registry and UniProt.
- Developed a ranking and filtering model improving alignment accuracy to >95% and increasing retrieval coverage by ~400%.

Student Research Internship | UCSB, Four "I"s lab Santa Barbara, CA | 06-08/2019

Supervisor: Prof. Matthew Turk and Prof. Tobies Hollerer

- Reimplemented a state-of-the-art pose tracking algorithm for local evaluation and deployment.
- Benchmarked performance against a novel method using standardized datasets.

PROFESSIONAL SERVICES & ACTIVITIES

Mentor, **EBRC Mentorship for Undergraduate and Master's Students Program (EMUMS)**	04/2025-Now
Direct Mentor, **Eric Xie**, Undergraduate Research Assistant, CIDAR Lab	06/2023-Now
International Workshop on Bio-Design Automation (IWBDA'25), **Oral Presentation**	09/2025
International Genetically Engineered Machine competition (iGEM'19, iGEM'24)	11/2019, 11/2024
Inventor/Contributor, **Patent: Agricultural Heavy Metal Biosensor-Integrated Device**	05-11/2024
Student Mentor of iGEM team: BostonU	04-10/2024
Chicago BioEngineering Conference (CBEC'24)	09/2024
Teaching Assistant, **EC504: Advanced Data Structures and Algorithms**, Boston University	01-05/2022, 01-05/2023
Engineering Biology Research Consortium (EBRC'23)	06/2023
International Workshop on Bio-Design Automation (IWBDA'22), Poster Presentation	10/2022

PUBLICATIONS

Google Scholar (https://scholar.google.com/citations?user=Zel-iSQAAAAJ&hl=en)

Journals:

[**J1**, IEEE-TCAD] **Zhou, Y.**, Oliveira, S. M., Sanka, R., McIntyre, D., & Densmore, D. (2024). Vespa: Logic-Level Constraint-Based Validation for Continuous-Flow Microfluidic Devices.

[**J2**, Nature Chemical Biology] Padmakumar, J. P., Sun, J. J., Cho, W., **Zhou, Y.**, Krenz, C., Han, W. Z., ... & Voigt, C. A. (2024). Partitioning of a 2-bit hash function across 66 communicating cells.

[**J3**, IEEE-TCBB] **Zhou, Y.**, Voigt, C. A., & Densmore, D. (2025). Constraint-Based Sub-Graph Partitioning for Multi-Cellular Biological Networks.

Conferences & Workshops:

[**C1**, IWBDA'22] A Conceptual Interactive Microfluidic Design and Control Workflow

[**C2**, IWBDA'25] LaNVis: Biological Constraint-based Large Network Visualizer (Accepted for oral presentation)

Manuscripts in Preparation:

[**J4**, in progress, targeting Nature Computational Science] *Neptune: Web-based Automated Design and Synthesis Platform for Microfluidic Devices.*